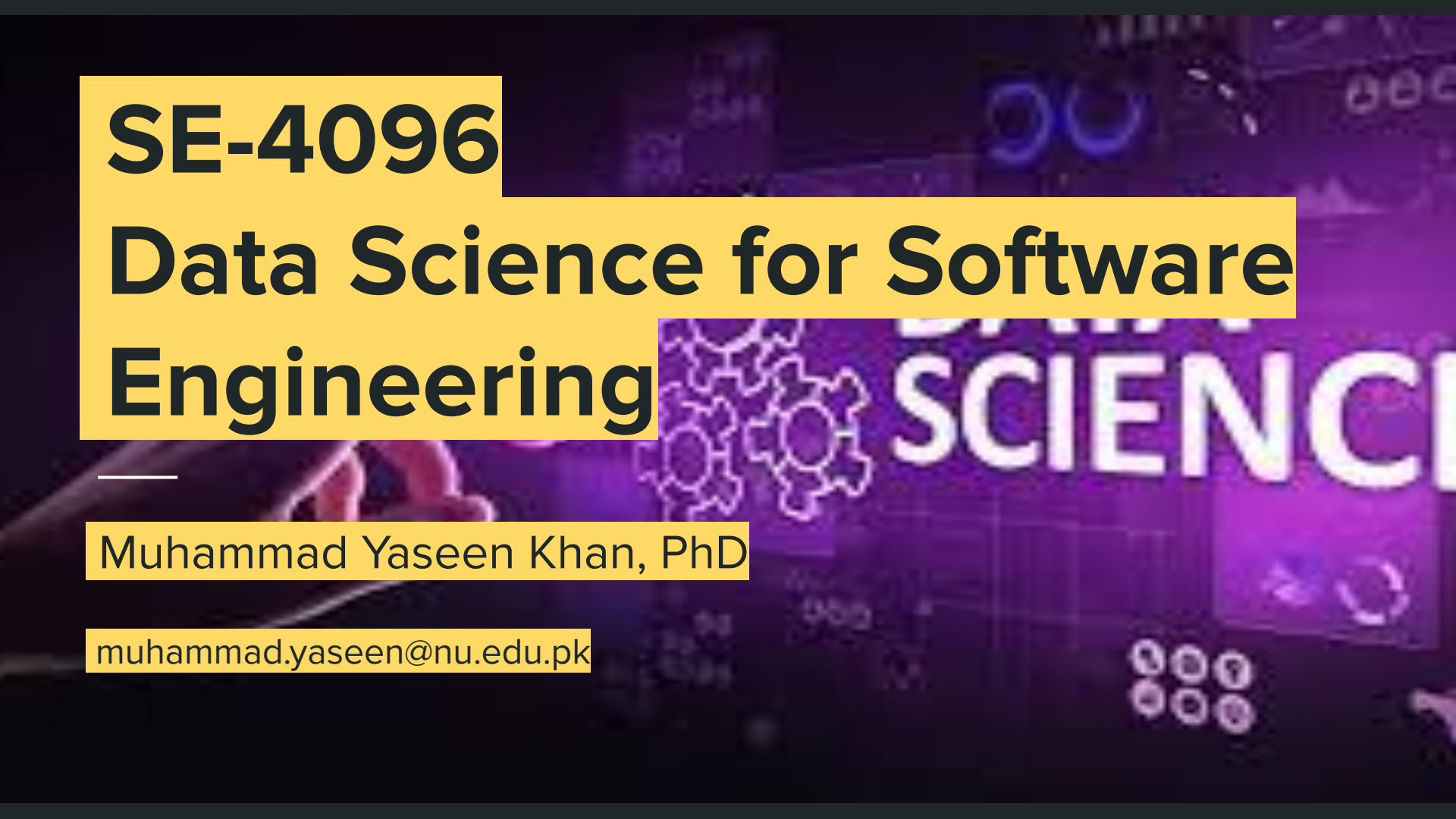




SE-4096

Data Science for Software Engineering

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Dealing with Classification Problem

Why Classification in Software Engineering?

Why Classification in Software Engineering?

In software projects we often want to predict:

1. Buggy vs Non-Buggy modules
2. High risk vs Low risk commits
3. Delay vs On-time tasks
4. Defect severity level

All of these are classification problems

There is a variety of classification algorithms that works with the probabilistic, entropy, linear/dichotomic space-division, neural network-inspired approaches for dealing the classification problem.

Machine Learning (basics)

Typically a mathematical representation of ML function is $f: X \rightarrow Y$

f is the function that is set to learn the predictive model using labelled data; where X is the set of input features, and Y is the output/target label

So in a dataset, we have to separate the target label and input features. Keeping the target in the data will definitely be cheating.

A good dataset is:

- Statistically show a balanced-distribution w.r.t the target outcome
- Representative
- Clean and unambiguous

Today We'll Learn how **Naïve Bayes Classifier** predicts such labels using probability.

Background: What is Probability?

Chance of something to happen...

Probability of event A:

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$$

Example:

- 20 modules
- 5 buggy

$$P(\text{Buggy}) = 5/20 = 0.25$$

Which means that the 25% modules are buggy

Background: Conditional Probability?

Chance of something to happen but depending on some condition/context...

Probability of event A:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Meaning: Probability of A given B has happened

Example: What is probability that module is buggy given churn is high?

$$P(Buggy|HighChurn)$$

Now prediction becomes context dependent

From Conditional Probability to Bayes Rule

Start with

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Also:

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Equate both:

$$P(A \cap B) = P(B|A)P(A)$$

Substitute: working on board

Bayes Theorm

Bayes Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Terms:

- $P(A)$ → Prior (Probability of event before seeing any evidence; background probability)
- $P(B|A)$ → Likelihood (Probability of observing evidence if A is true; How likely is the data given the class)
- $P(B)$ → Evidence (Overall probability of the evidence; How common is this feature in entire dataset)
- $P(A|B)$ → Posterior (Final Prediction)

Classification Using Bayes

For classification

$$P(\textit{Class}|\textit{Features})$$

Example

$$P(\textit{Buggy}|\textit{Churn}, \textit{LOC}, \textit{Developers})$$

We will compute for both class labels (buggy and not-buggy) and choose the class with the highest probability.

Typically it is represented with `argmax`

Defect Prediction Dataset

Module	LOC	Churn	Devs	Complexity	Buggy
M1	High	High	3	High	Yes
M2	Low	Low	1	Low	No
M3	Medium	High	2	Medium	Yes
M4	Low	Low	1	Low	No
M5	High	Medium	4	High	Yes
M6	Medium	Low	2	Low	No
M7	High	High	3	High	Yes
M8	Low	Medium	1	Medium	No

Problem with the Bayes

Direct computations $P(Buggy|x_1, x_2, x_3, \dots, x_n)$

This needs to have data like $P(x_1, x_2, x_3, \dots|Buggy)$

At backend:

- Too complex to be existing
- Too many combinations
- Requires huge data

Naïve Assumption

Key Idea: Assume features are independent

$$P(x_1, x_2, x_3|C) = P(x_1|C)P(x_2|C)P(x_3|C)$$

Where the features, instead of a joint record, are treated separately. Hence, we say that the features are mutually independent. this is called Naïve Independence Assumption, which simplifies computation drastically.

Final Classifier

$$P(C|X) \propto P(C) \prod_{i=1}^n P(x_i|C)$$

Note, since we are to multiply a lot of probabilities [0,1] therefore, we will definitely undergo with the smaller numbers in the production. Hence, a remedy is to:

- apply log on both sides, that will eventually convert production in summation
- Use Laplace Smoothing by adding 1 in the counts

Types of Naïve Bayes Algorithm

Gaussian NB → when features are continuous values

$$P(C|X) \propto P(C) \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma_{Ci}^2}} \exp\left(-\frac{(x_i - \mu_{Ci})^2}{2\sigma_{Ci}^2}\right)$$

Multinomial NB → when features are counts/text

$$P(C|X) \propto P(C) \prod_{i=1}^n P(x_i|C)$$

Bernoulli NB → when features are binary

$$P(C|X) \propto P(C) \prod_{i=1}^n p_{ic}^{x_i} (1 - p_{ic})^{(1-x_i)}$$

Hands-on Classification with Naïve Bayes

Module	LOC	Churn	Devs	Complexity	Buggy
M10	High	Low	3	Low	?

Lab:

- Introduction to
 - Train Test Splits
 - LOO, x-validation
 - Naive Bayes with scikit-learn
 - Evaluation Metrics